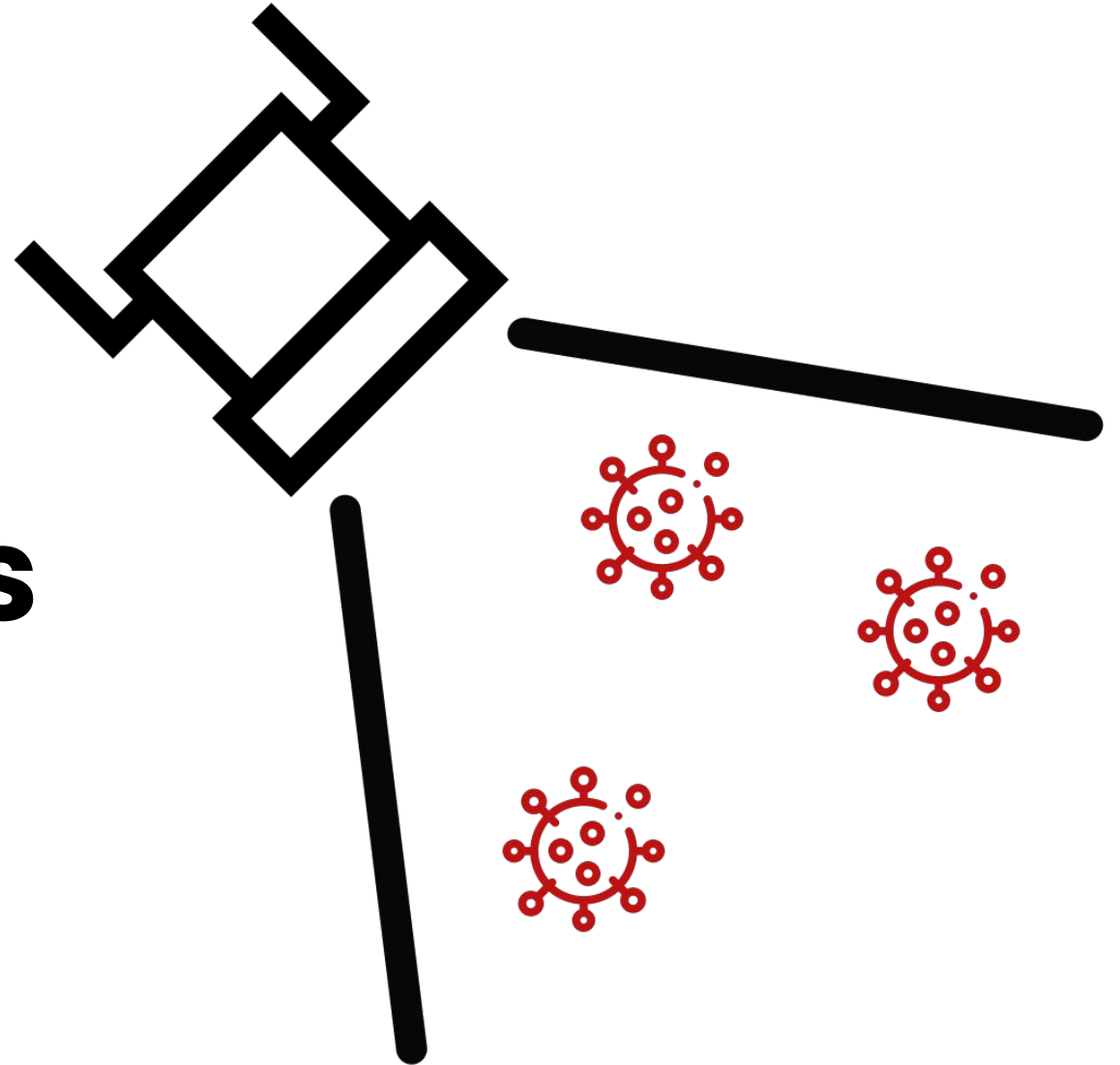


FloodLAMP Biotechnologies

A Public Benefit Corporation

Randy True

randy@floodlamp.bio



The COVID crisis in a nutshell...



**Bloomberg
Businessweek**

October 5, 2020

Want to go to a restaurant?

We need better testing

A sports event? A movie?

We need better testing

Travel somewhere? Anywhere?

We need better testing

Go to a wedding? A funeral? A party?

We need better testing

Send your kids back to an actual school?

We need better testing

A normal-ish life before there's a vaccine?

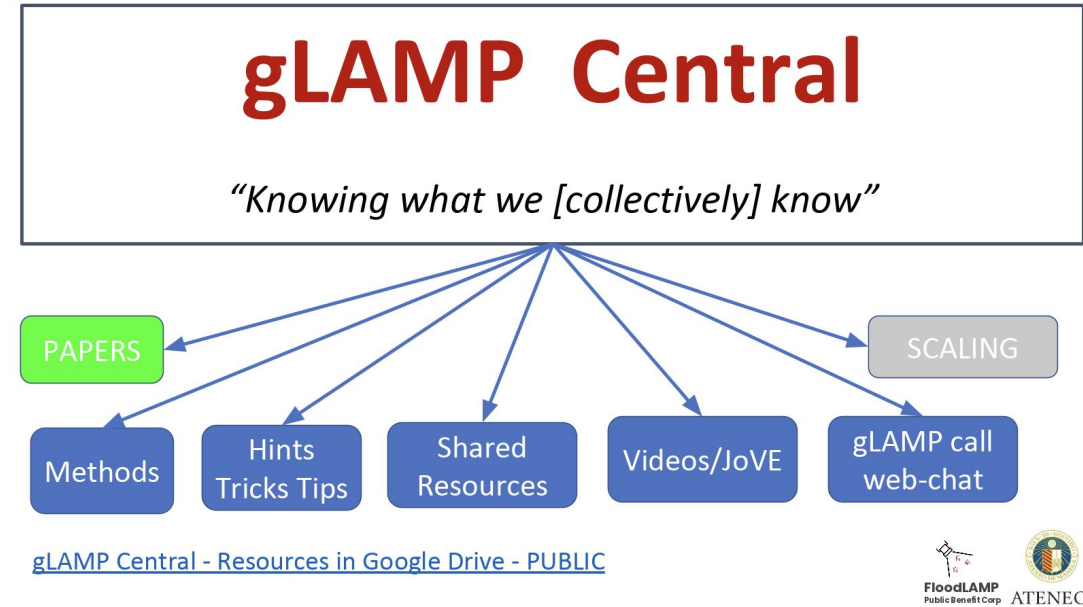
We need better testing

**Our Mission is
to develop mass
disease screening and
usher in a new era of
Open Diagnostics**

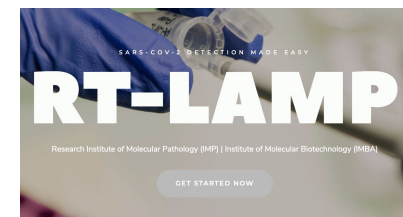


Why Open Protocols?

- Open protocols will have maximum impact on increasing testing capacity and lowering cost
- Allows for cross functional sharing of best practices
- Removes dis-incentives which have hampered industry during the crisis
- Strategy for priority at FDA and expedited EUA authorization



**OPEN
COVID
PLEDGE**



Vienna
BioCenter

Who We Are



Randy True Founder & CEO

- Founder TMI Inc. - Acq. Affymetrix 25MM
- VP of R&D at Affymetrix;
- Founder 501(c)3 STEM Education Nonprofit
- CTE Teaching Credential
- Stanford, B.S. Physics, M.S. EE

**Proven Biotech Entrepreneurial
& Science Education Nonprofit
Experience**

FloodLAMP Team

Theresa Ling UX Design, Strategy, Communications

- Former Design Lead at Uber and New Relic

Carey Tan Director of Operations and Logistics

Antanas Sadunas Lab Technician, All Around Fixer

Sam Fogelberg Lab Integration Engineer

Scientific Advisory Board

Anne Wyllie, Ph.D.

- Associate Research Scientist at Yale School of Medicine

Bill Hyun, MS

- Clinical Director at UC San Francisco

Connie Cepko, Ph.D.

- Bullard Professor of Genetics and Neuroscience, Harvard Medical School, HHMI

Ned Augenblick, Ph.D.

- Associate Professor, Economic Analysis and Policy Group at UC Berkeley Haas School of Business

Advisors

Tim Lugo Advisor, Strategic Planning and Outreach

- Biotechnology Group Head at William Blair

Zarak Khurshid Advisor, Modeling and Industry Intel

- Asymmetry Capital, former MDx Analyst at Wedbush

John Edge

- Oxford Internet Institute, Blue Field Labs, Broadhaven Capital Partners, ID2020



What is a Public Benefit Corporation?

Well Defined Mission

- Explicit in our Founding Documents
- Provides North Star to navigate critical decisions

Different Fiduciary Standard

- Mission has real ramifications on how our organization is run
- Maximizing public good is the prime directive, not maximizing shareholder value
- Board must consider both and liability is shielded
- Board decisions must be disinterested and not involve any conflict of interest

Transparency

- PBCs must report on how it is promoting and advancing its public benefit purpose
- FloodLAMP is living up to a commitment to very high transparency already

<https://www.hutchlaw.com/blog/whats-the-difference-between-a-public-benefit-corporation-and-a-b-corp-certific>

FloodLAMP.bio



contact@floodlamp.bio

State of Delaware
Secretary of State
Division of Corporations
Delivered 10:02 AM 08/03/2020
FILED 10:02 AM 08/03/2020
SR 20206538043 - File Number 3362851

CERTIFICATE OF INCORPORATION

OF

FLOODLAMP BIOTECHNOLOGIES, PBC

A PUBLIC BENEFIT CORPORATION

ARTICLE I

The name of the corporation is FloodLAMP Biotechnologies, PBC (the “Corporation”).

ARTICLE II

The address of the Corporation’s registered office in the state of Delaware is 1209 Orange Street, in the city of Wilmington, county of New Castle, Zip Code 19801. The name of its registered agent at such address is The Corporation Trust Company.

ARTICLE III

The specific public benefit that the Corporation will promote is the development of open protocols and the implementation of large scale screening for infectious disease for the benefit of the public. In addition, the Corporation may engage in any lawful act or activity for which corporations may be organized under the Delaware General Corporation Law as the same exists or may hereafter be amended (the “DGCL”).

Open COVID Pledge



Immediate action is required to halt the COVID-19 Pandemic and treat those it has affected. It is a practical and moral imperative that every tool we have at our disposal be applied to develop and deploy technologies on a massive scale without impediment.

We therefore pledge to make our intellectual property available free of charge for use in ending the COVID-19 pandemic and minimizing the impact of the disease.

We will implement this pledge through a license that details the terms and conditions under which our intellectual property is made available.

opencovidpledge.org

nature biotechnology

Explore our content ▾

Journal information ▾

[nature](#) > [nature biotechnology](#) > [patents](#) > [article](#)

Patents | [Published: 05 October 2020](#)

PATENTS

Pledging intellectual property for COVID-19

[Jorge L. Contreras](#) , [Michael Eisen](#), [Ariel Ganz](#), [Mark Lemley](#), [Jenny Molloy](#), [Diane M. Peters](#) & [Frank Tietze](#)

Nature Biotechnology **38**, 1146–1149(2020) | [Cite this article](#)

3133 Accesses | **70** Altmetric | [Metrics](#)

Voluntary pledges to make intellectual property broadly available to address urgent public health crises can overcome administrative and legal hurdles faced by more elaborate legal arrangements such as patent pools and achieve greater acceptance than governmental compulsory licensing.

www.nature.com/articles/s41587-020-0682-1

FloodLAMP's 3 Major Initiatives

- 1) Ultra-low-cost, instrument-free molecular test
- 2) Distributed sample pooling program
- 3) Open Source EUA Consortium

U.S. Regulatory Pathways to COVID Testing

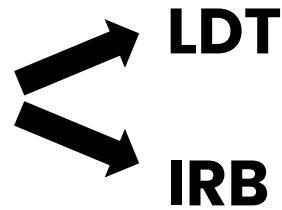


1) EUA

- Dx testing at CLIA labs
- LDT vs IVD
- Complexity (High, Med)
- Designation process
- CLIA Waived (POC)
- ROR (Right of Reference)



2) CLIA LAB



Any Lab*



3) "SURVEILLANCE"

- no EUA, no CLIA
- "non-CLIA individual results"
- refer "SARS-CoV-2 detected" to diagnostic test



CDC ROR for PCR primers – LAMP next

Immediate, practical,
impactful first step

Suggest 2 of As1e,N2,E1

Partner for commercial
product

*keep cost as low as possible

Saturday, October 31st ▾

Randy 1:51 PM

Any test developer requesting an EUA that intends to leverage data from another EUA-authorized device must obtain a right of reference to leverage the performance data for that EUA-authorized device. The Centers for Disease Control and Prevention (CDC) has granted a right of reference to the performance data contained in CDC's EUA request (FDA submission number EUA200001) to any entity seeking an FDA EUA for a COVID-19 diagnostic device.



<https://www.fda.gov/medical-devices/coronavirus-disease-2019-covid-19-emergency-use-authorizations-medical-devices/vitro-diagnostics-euas>

 **U.S. Food and Drug Administration**

In Vitro Diagnostics EUAs

In Vitro Diagnostics EUAs for COVID-19 Tests (672 kB) ▾



"FDA does not authorize EUAs for general methods. EUAs are authorized for "test systems," where a specific set of reagents/instruments are used according to a particular testing protocol for a SARS-CoV-2 test. In other words, the Saliva Direct method would only be authorized using the instruments/reagents and SARS-CoV-2 test for which you have submitted validation data in your EUA."

👍 1 🗨️



Unlocking Regulatory Path with Open EUAs



Open EUA Consortium

SalivaDirect

**FloodLAMP
(Rabe Cepko)**

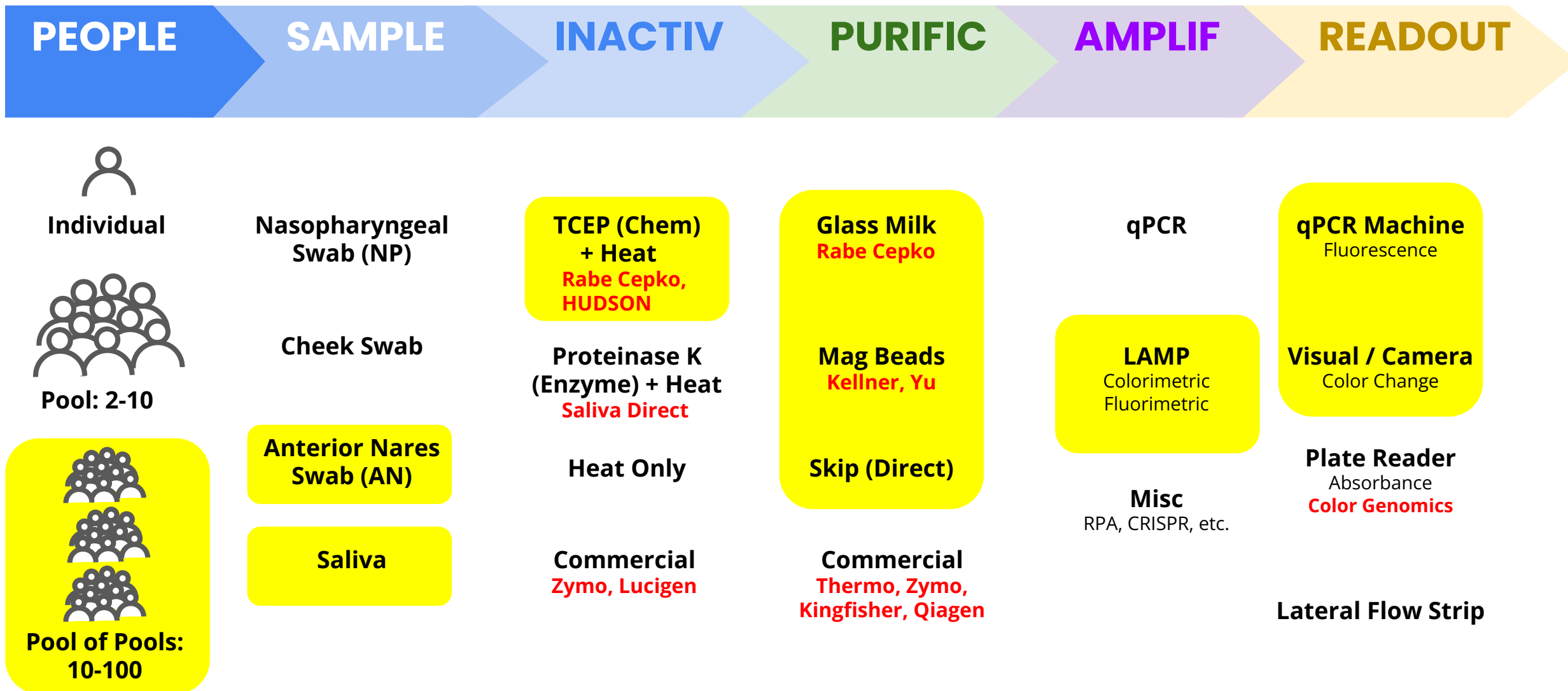
Direct LAMP

Pooling

Automation



Molecular Assay Landscape



Molecular Assay Subway Map

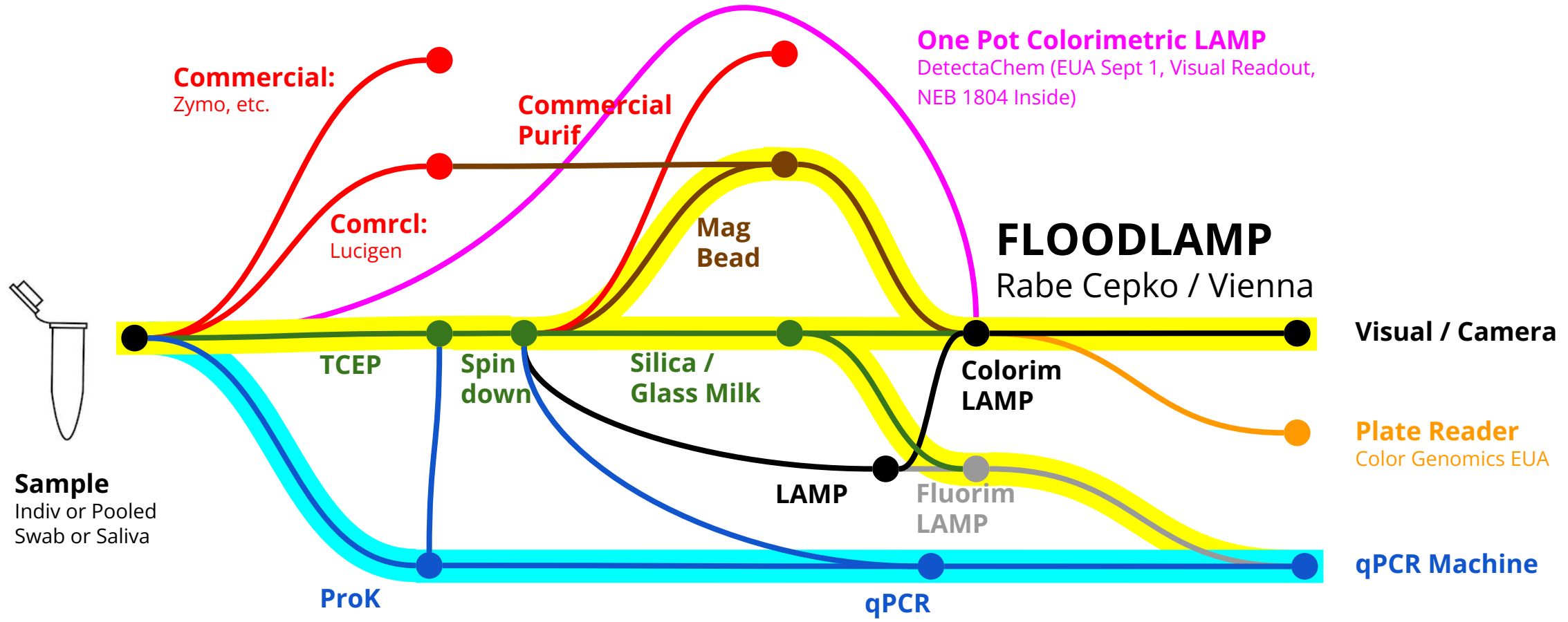
SAMPLE

INACTIV

PURIFIC

AMPLIF

READOUT



SALIVA DIRECT



Thanks!

Recent LAMP IVD EUA at 75cp/ml

1st Colorimetric RT-LAMP EUA
Small company from Texas

MOBILEDETECT BIO



- From Detectachem
- cLAMP, heater+smartphone app
- EUA for swabs in CLIA, MD heater or variety of others
- N+E primers
- LOD: 75 copies/ μ L

DETECTACHEM MOBILEDETECT BIO

MobileDetect-BIO BCC19 Test Kit
for SARS-CoV-2 Detection

Instructions for Use – Ver 1.8

September 1, 2020

For *In Vitro* Diagnostic Use Only. **IVD**
Prescription Use only. **Rx ONLY**
For Emergency Use Authorization only.

SUMMARY AND EXPLANATION

An outbreak of respiratory illness of unknown etiology in Wuhan City, Hubei Province, China was initially reported to the World Health Organization (WHO) on December 31, 2019. Chinese authorities identified a novel coronavirus (2019-nCoV) which has resulted in thousands of confirmed human infections in multiple provinces throughout China and exported cases in several Southeast Asian countries and more recently the United States. Cases of severe illness and some deaths have been reported. The International Committee for Taxonomy of Viruses (ICTV) renamed the virus SARS-CoV-2.



LoD Targets Changing

COVID-19 Update: FDA Publishes Comparative Performance Data for COVID-19 Molecular Diagnostic Tests

U.S. Food and Drug Administration sent this bulletin at 09/16/2020 09:03 AM EDT

If your email program has trouble displaying this email, [view as a webpage](#).



FDA Publishes Comparative Performance Data for COVID-19 Molecular Diagnostic Tests

On Tuesday, September 15, 2020, the U.S. Food and Drug Administration published comparative performance data for some authorized COVID-19 molecular diagnostic tests. The tables show the Limit of Detection (LoD) of more than 55 authorized molecular diagnostic COVID-19 tests against a standardized sample panel provided by the FDA.

The FDA provided these standardized samples, known as a **reference panel**, to test developers who are required to assess their test's performance against this panel (or other FDA-recommended reference materials) as a condition of their Emergency Use Authorization (EUA).

The data published today gives laboratories, health care providers, and patients a new resource on the relative performance of available tests to better inform which tests they choose to use. The FDA will continue to update the table as it receives additional results.

[See Reference Panel](#)

Questions?

Email CDRH-EUA-Templates@fda.hhs.gov.

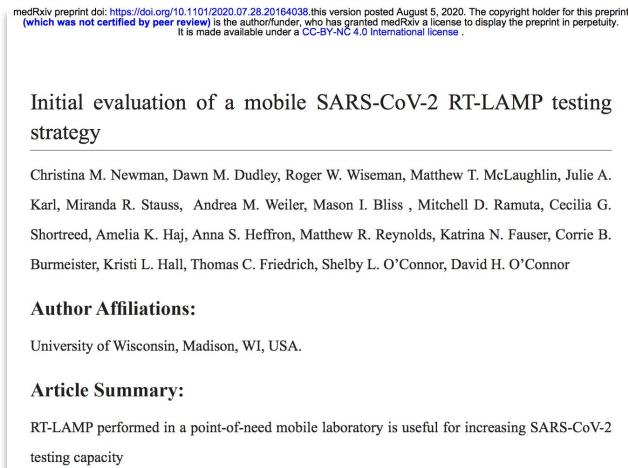
FDA Comparative dataset suggests LoD of leading tests 10x-40x different than published results

Copy of FDA SARS-CoV-2 Reference Panel Comparative Data					
File Edit View Insert Format Data Tools Add-ons Help Last edit was yesterday					
100% \$ % .00 .00 123 Default (Arial) 10 B I					
	A	B	C	D	E
1	Test	Developer	EUA LOD (/3ml swab)	FDA Panel LOD (/3ml swab)	FDA/EUA LOD
2	Influenza SARS-CoV-2 (Flu SC2) Multiplex Assay	Centers for Disease Control and Prevention (CDC)	TCID	17,100	?
3	ARIES SARS-CoV-2 Assay	Luminex Corporation	999	540,000	540.5
4	QIAstat-Dx Respiratory SARS-CoV-2 Panel	QIAGEN GmbH	1,500	540,000	360
5	SARS-CoV-2 PCR Test	Diatherix Eurofins Laboratory	3,000	540,000	180
6	Abbott RealTime SARS-CoV-2 assay	Abbott Molecular	120	16,200	135
7	BD SARS-CoV-2 Reagents for BD MAX System	Becton, Dickinson & Company	120	16,200	135
8	AMPIPROBE SARS-CoV-2 Test System	Enzo Life Sciences, Inc.	840	54,000	64.3
9	BioGX SARS-CoV-2 Reagents for BD MAX System	Becton, Dickinson & Company (BD)	120	5,400	45
10	cobas SARS-CoV-2	Roche Molecular Systems, Inc. (RMS)	138	5,400	39.1
11	LetsGetChecked Coronavirus (COVID-19) Test	PrivaPath Diagnostics, Inc.****	600	21,600	36
12	Color Genomics SARS-CoV-2 RT-LAMP Diagnostic	Color Genomics, Inc.****	2,250	54,000	24
13	Boston Heart COVID-19 RT-PCR Test	Boston Heart Diagnostics	750	16,200	21.6
14	Xpert Xpress SARS-CoV-2 test	Cepheid	750	16,200	21.6
15	PerkinElmer New Coronavirus Nucleic Acid Detection	PerkinElmer, Inc.	28	540	19.3
16	ID NOW COVID-19	Abbott Diagnostics Scarborough, Inc.	313	6,000	19.2
17	Lilly SARS-CoV-2 Assay	Eli Lilly and Company	3,000	54,000	18
18	Psoma COVID-19 RT Test	Psomagen, Inc.	3,000	54,000	18
19	BioFire COVID-19 Test	BioFire Defense, LLC	990	16,200	16.4
20	Quest SARS-CoV-2 rRT-PCR	Quest Diagnostics Infectious Disease, Inc.	408	5,400	13.2
21	CareStart COVID-19 MDx RT-PCR	Access Bio, Inc.	1,286	16,200	12.6
22	EURORealTime SARS-Cov-2	Euroimmun US, Inc.	450	5,400	12



"Capable to Infect Others"

Academics such as Mina, O'Connor, Mason have been leading the charge since early August



We obtained Institutional Review Board approval (Protocol #2020-0855) to prototype a simple, mobile, non-diagnostic SARS-CoV-2 testing workflow. Replication-competent SARS-CoV-2 is rarely recovered from individuals whose nasopharyngeal swab viral loads are less 1E+6 copies / mL, while peak respiratory viral loads exceeding 1E+10 copies / mL have been observed [3]. Furthermore, detection of virus in saliva is correlated with presence of virus in nasopharyngeal swab samples [4]. Therefore, we reasoned that a cost-effective testing strategy that detects at least 1E+6 copies / mL in saliva and provides near real-time results could be effective in identifying those at greatest risk of infecting others, encouraging isolation and follow-up diagnostic testing.

Adoption of 1e5 copies/ml is emerging as consensus target.

Screenshot from a CDC Working Group

Viral Load in Saliva (virus copies/ml)	PCR Test (Gold standard)	Home Test (Less sensitive)	Capable to Infect Others?
1,000	+	-	No
100,000	+	+	No
1,000,000	+	+	Yes

Saliva Direct "most unique EUA" – FDA

- Is a protocol not a product
- Yale can "designate" CLIA labs in streamlined process compared to LDT

- Yale team constrained by resources (50 and counting of 400 labs designated)

The screenshot shows a webpage from the Yale School of Medicine. The header includes the Yale logo and navigation links: About YSM, Education, Research, Patient Care, Diversity, News, and A-Z Index. The main title is "SalivaDirect: What You Need to Know About the New COVID-19 Test", dated September 03, 2020, by Michael Greenwood. Below the title is a social media sharing bar. The main content area features a diagram of the SalivaDirect test process: a person provides saliva, which is combined with Proteinase K, then heat inactivated at 95°C, and finally analyzed using Dualplex RT-qPCR. A sidebar on the left contains a "COVIDTracker" logo and a link to "About our Test Development". A bottom sidebar notes that SalivaDirect has moved to a new page and provides a link to the new page.

SalivaDirect: Lab authorization request form

SalivaDirect received Emergency Use Authorization (EUA) from the Food and Drug Administration (FDA) on August 15th, 2020.

The SalivaDirect FDA EUA is for a Laboratory Developed Test, but we have the right to designate other labs its use.

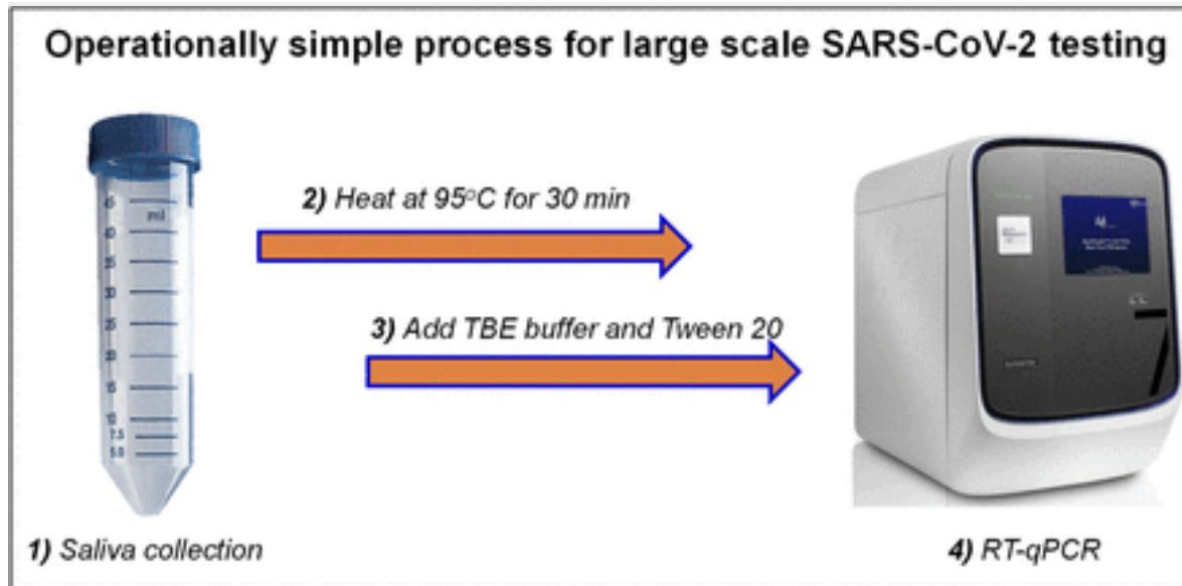
Only high complexity CLIA-certified labs in the United States can become authorized to run SalivaDirect. Please fill out this form if you would like to receive more information on how to become a designated lab.

Do you have access to at least 5x SARS-CoV-2 positive and 5x negative saliva samples for CLIA verification? *

Would you be open to collaboration on bridging studies for your automation systems?

SHIELD – Univ of Illinois

- Very similar to Yale's Saliva Direct



 UNIVERSITY OF ILLINOIS SYSTEM

AboutDataDirectoryInitiativesSystem Offices

BENEFITS OF OUR TEST

	LOW COST	The SHIELD Illinois test costs \$20-\$30, compared to \$100+ for a standard nasal swab test
	FAST NOTIFICATION	Current notification time for the saliva test is only 6-12 hours compared to a 2-3 day turnaround time for most currently available tests.
	WIDESPREAD TESTING	Through advantages in cost, speed and accuracy, the saliva tests can increase access to testing and control spread by identifying pre-symptomatic and asymptomatic carriers.
	EASE OF USE	The saliva-based test is non-invasive and does not need medically trained personnel to collect samples.
	HIGH ACCURACY	The saliva test has a specificity of 99.8-99.9%, with very few false positives. High specificity combined with frequent testing makes the SHIELD saliva-based test extremely accurate.

EXPANSION IN ILLINOIS

SHIELD Illinois is partnering with universities, companies, school districts, healthcare facilities, and other organizations to launch collection sites. Any organization can become a testing sponsor by subsidizing the cost of tests for a group of people.

RESPONSIBILITIES		
TEST SPONSOR	COLLECTION SITE PARTNER	LAB PARTNER*
• Commit to paying for a minimum number of tests	• Provide space to collect up to 1,000+ samples per	• Manage, staff, and run the testing in a CLIA-certified



LAMP EUAs for SARS-CoV2

EUA Date	Test	Diagnostic	Target	IC	Detection	Extraction	Amplification	LOD spike	LOD GE/3ml swab
Nov-17	Lucira (At Home POC)	Lucira COVID-19 All-In-One Test Kit	N, N	External + IC	Colorimetric	~Direct	Lucira device	Inactivated virus	2,700
Oct-05	Seasun (2 duplex)	AQ-TOP COVID-19 Rapid Detection Kit PLUS	orf1ab, N	Human RNaseP	PNA Probe	Manual (Qiagen_60704 or Seasun_SS-1300) or Automated (Panagene_PNAK-1001 on PanaMax48)	BioRad_CFX96 or ABI7500	NCCP_43326 genomic RNA	3000
Sep-01	Detectachem	MobileDetect Bio BCC19 (MD-Bio BCC19) Test Kit	N, E	only external controls	Colorimetric	Direct (1µl of transport media into LAMP)	heat block or qPCR (MD-Bio heater or BioRad_T100 or AB_Veriti or ...)	Twist_MT007544.1 synthetic gRNA	225,000
Aug-31	Mammoth	SARS-CoV-2 DETECTR Reagent Kit	N	Human RNaseP	CRISPR Probe	Automated (Qiagen_955134 on Qiagen_EZ1AdvancedBenchtop)	ABI7500	SeraCare_AccuPlex_0505-0168 IVT encapsulated RNA	60,000
Aug-13	Pro-Lab	Pro-AmpRT SARS-CoV-2 Test	RdRP	only external controls	Probe	Direct (swab into 0.1ml) or kit (swab into 1ml then Pro-Lab_PLM-2000)	Optigene_GenieHT	BEI_NR-52287 inactivated virus	125*
Jul-09	UCSF/Mammoth	SARS-CoV-2 RNA DETECTR Assay	N	Human RNaseP	CRISPR Probe	automated (Qiagen_955134 on Qiagen_EZ1)	ABI7500	SeraCare_AccuPlex_0505-0126 (lot 10480311) IVT RNA	60,000
May-21	Seasun (1 duplex)	AQ-TOP COVID-19 Rapid Detection Kit	orf1ab	Human RNaseP	PNA Probe	manual (Qiagen_60704)	BioRad_CFX96 or ABI7500	NCCP_43326 gRNA	21,000
May-18/ Aug31	Color	Color Genomics SARS-CoV-2 RT-LAMP Diagnostic Assay	N, E, nsp3	Human RNaseP	Colorimetric	automated (PerkinElmer_CMG-1033 on PerkinElmer_Chemagic360)	plate reader (Biotek_NEO2), Hamilton_Star	ATCC_VR-1986D (gRNA)	2,250
May-06	Sherlock BioSci	Sherlock CRISPR SARS-CoV-2 Kit	orf1ab, N	Human RNaseP	CRISPR Probe	Manual (ThermoFisher_12280050)	heat block or qPCR, Plate reader (BioTek_NEO2)	gRNA	20,250
Apr-10	Atila BioSystems	iAMP COVID-19 Detection Kit	orf1ab, N	human GAPDH	Probe	Direct (swab into 350µl, 15min RT then 3µl to LAMP)	BioRad_CFX96 or ABI7500 or Roche_LightCycler480II or Atila_PG9600	SeraCare_AccuPlex_0505-0129 pseudovirus (recombinant alphavirus)	3,500*

from Matt McFarlane
(twitter @mattmcfar)

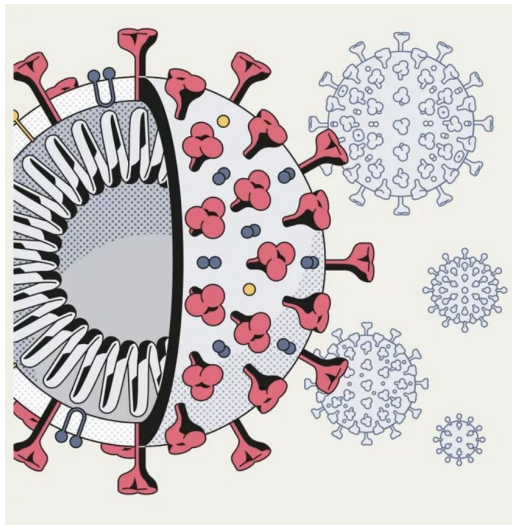


Ultra Low Cost Sample Prep

STEP 1:

Inactivation

- Breaks open virus and cells
- Preserves RNA
- Makes sample safe and improves results



credit: OpenCOVIDScreen

Rabe B, Cepko C. SARS-CoV-2 SARS-CoV-2 detection using isothermal amplification and a rapid, inexpensive protocol for sample inactivation and purification. PNAS 7-8-2020. <https://www.pnas.org/content/pnas/117/39/24450.full.pdf>

STEP 2:

Purification / Concentration

- Gives high sensitivity for pooling
- Magnetic option for automation-friendly workflow



Commercial products
(~\$5/sample)

Bulk form (silica)
(<\$0.01/sample)

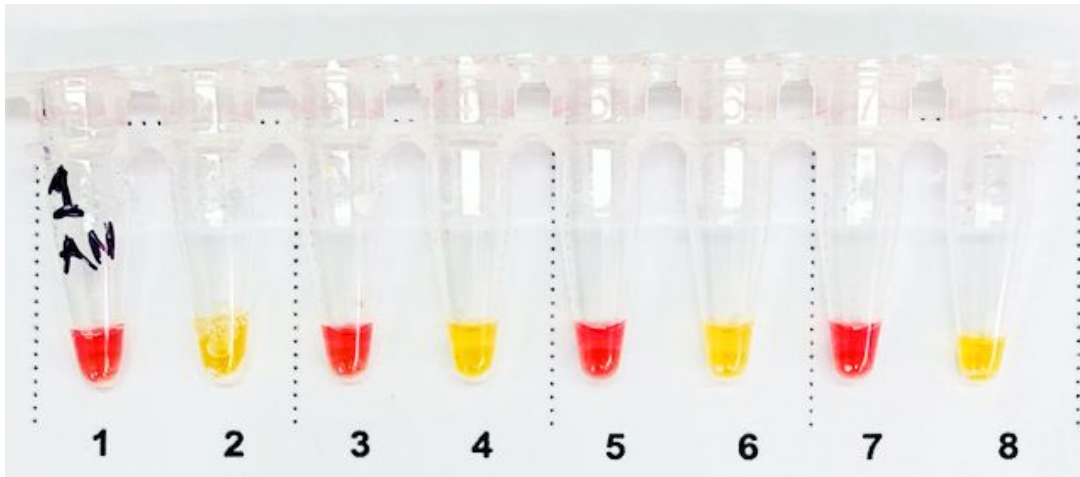
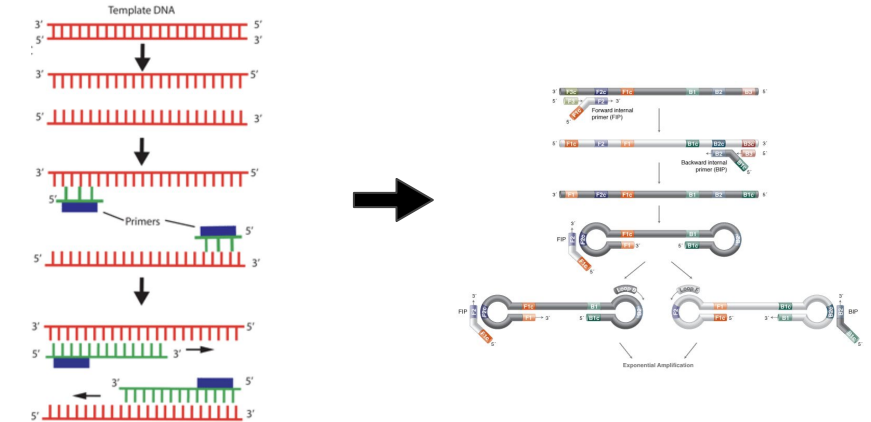


LAMP – Isothermal

STEP 3:

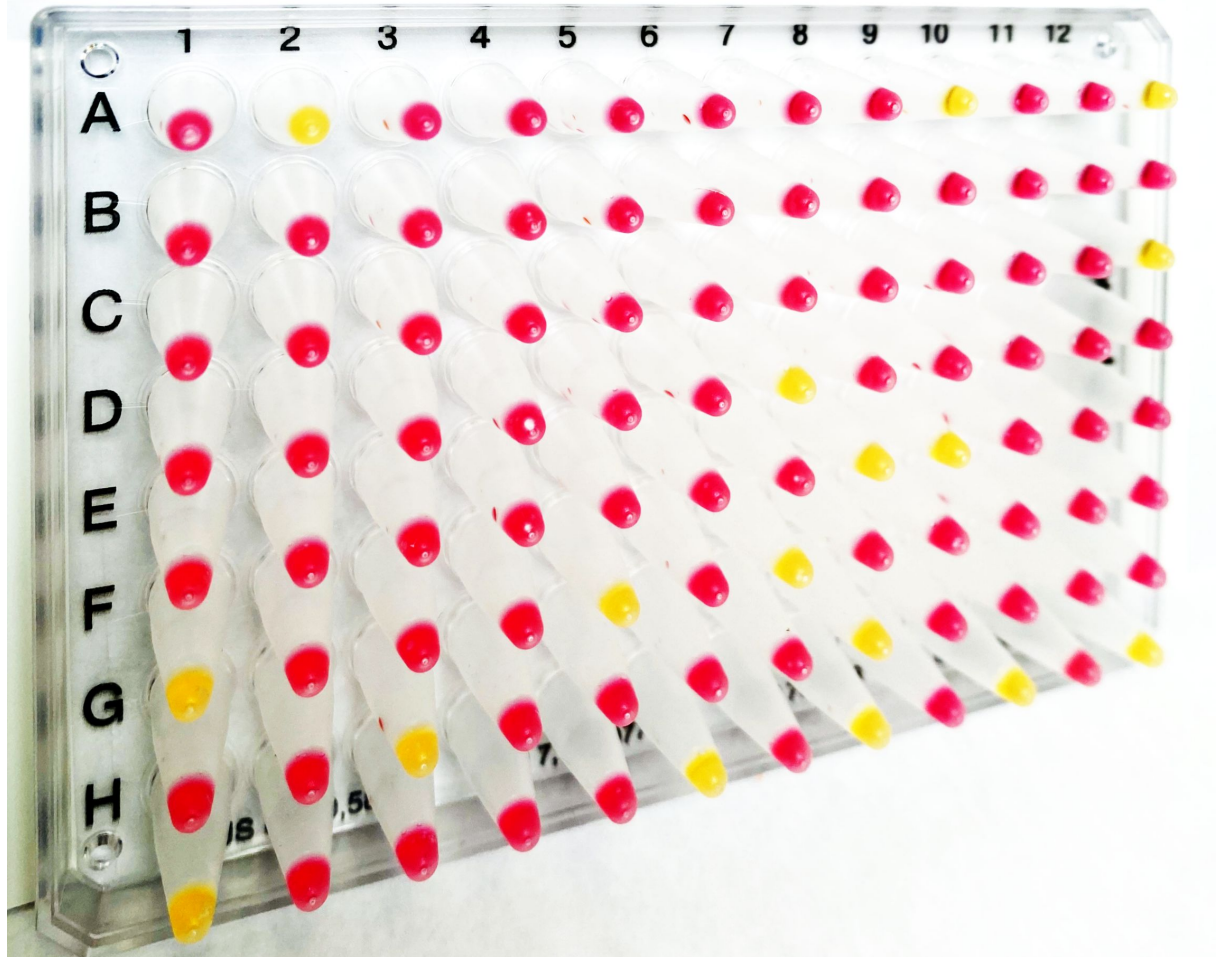
Amplification

- Detects viral RNA
- LAMP generates 10X more DNA in 1/3 time compared to PCR
- Easy visual / camera-based readout



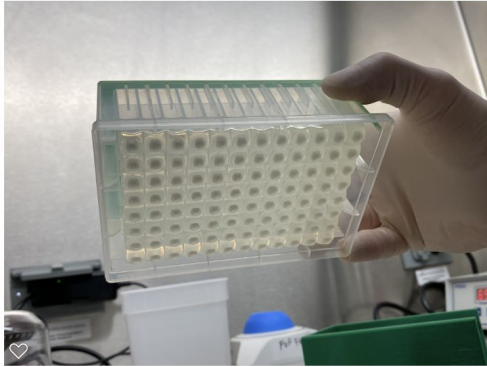
	qPCR	LAMP
Equipment	qPCR machine (thermal cycler & optics)	simple heat block (isothermal)
Time	90 min	30 min
Capital Cost	\$30K-100K	\$300

Colorimetric LAMP – Instrument Free

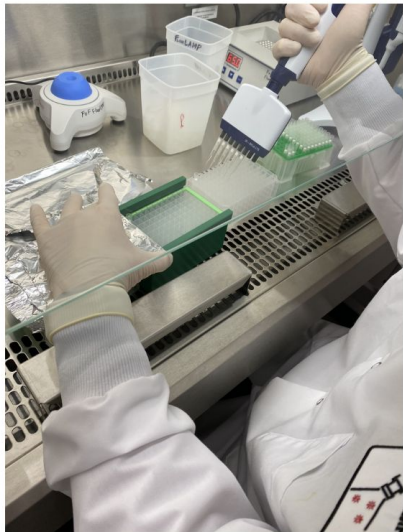


- 1hr hands-on time per tech
- 96 wells x 24 samples/pool = 2304 samples / plate
- Running more plates just requires more techs

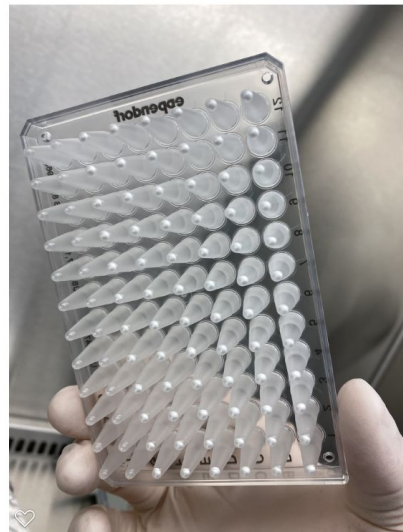
Instrument Free = Unlimited Scaling



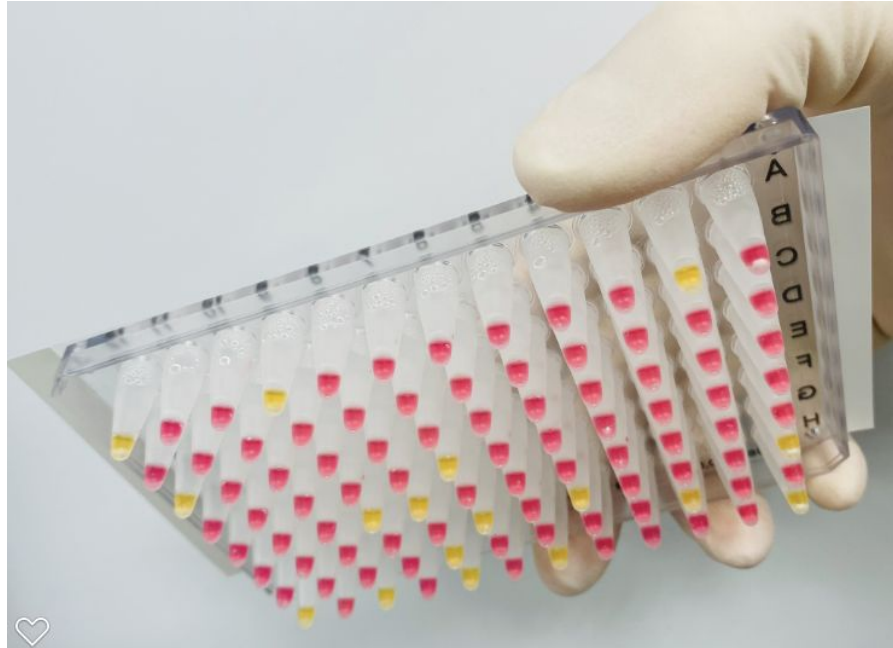
Silica pellets in deepwell plate after binding during purification step



3D Printed Guard for Multichannel aspiration



dried pellets - ready for LAMP ALL MM addition



FloodLAMP Data – LoD 2cp/ul

- High sensitivity, Low LoD of 2 copies/μl enables pooling to high levels (50-100)
- Realistic contrived positives: nasal swabs spiked with heat-treated virions before inactivation

9/22/2020 3:00 PM		Run IP8	Nasal Zepto Raw Spiked - 2 cp/ul			19/19	Negatives	4/4							
	1	2	3	4	5	6	7	8							
1st Strip	NZ 2cp/ul	NZ 2cp/ul	NZ 2cp/ul	NZ 2cp/ul	NZ 2cp/ul extra	NZ 2cp/ul	PI	PI*							
	9	10	11	12	13	14	15	16							
2nd Strip	NZ 2cp/ul	PI	NZ 2cp/ul	NZ 2cp/ul	UN	NZ 2cp/ul	NZ 2cp/ul	NZ 2cp/ul							
	17	18	19	20	21	22	23	24							
3rd Strip	NZ 2cp/ul	NZ 2cp/ul	UN	NZ 2cp/ul	NZ 2cp/ul	NZ 2cp/ul	NZ 2cp/ul	NZ 2cp/ul							
NZ 2cp/ul	19	<table><tr><th>Concentration (copies/μL)</th><th>Positive Samples Detected</th></tr><tr><td>4</td><td>100% (22/22)</td></tr><tr><td>2</td><td>100% (19/19)</td></tr><tr><td>1</td><td>55% (11/20)</td></tr></table>			Concentration (copies/μL)	Positive Samples Detected	4	100% (22/22)	2	100% (19/19)	1	55% (11/20)			
Concentration (copies/μL)	Positive Samples Detected														
4	100% (22/22)														
2	100% (19/19)														
1	55% (11/20)														
N UN	2														
PI (NTC)	2														
PI* (RNA PC)	1														
Summary: ALL CORRECT															



FloodLAMP Data – Clinical Positive Sample



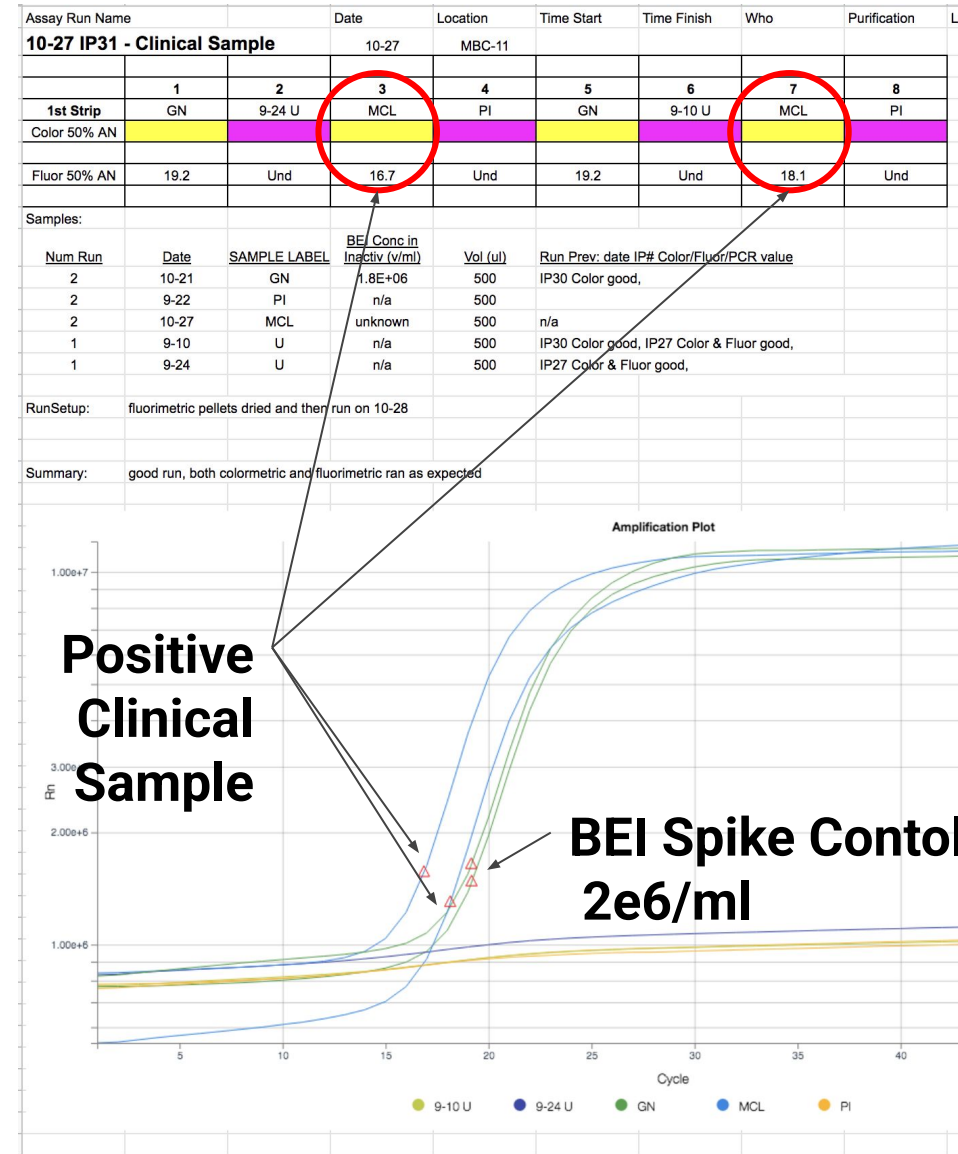
Colorimetric LAMP

- Visual / Camera Readout
- Instrument Free
- Practically Unlimited Scaling

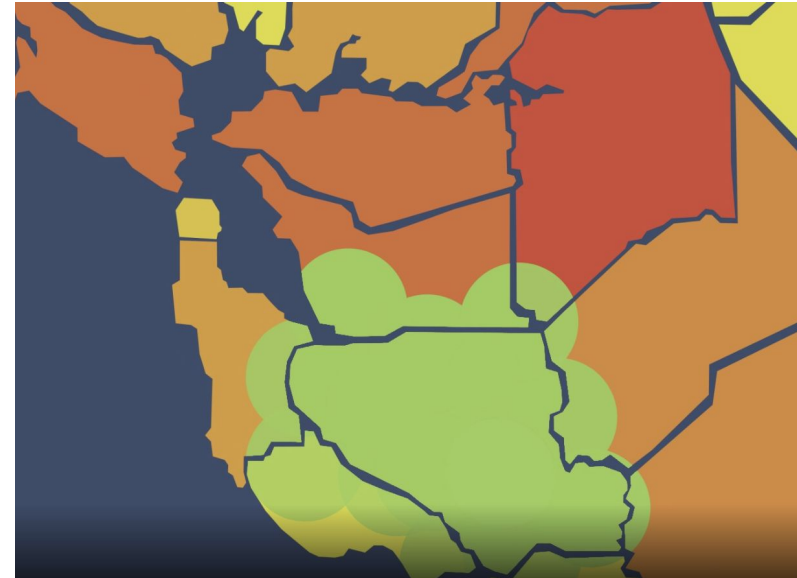
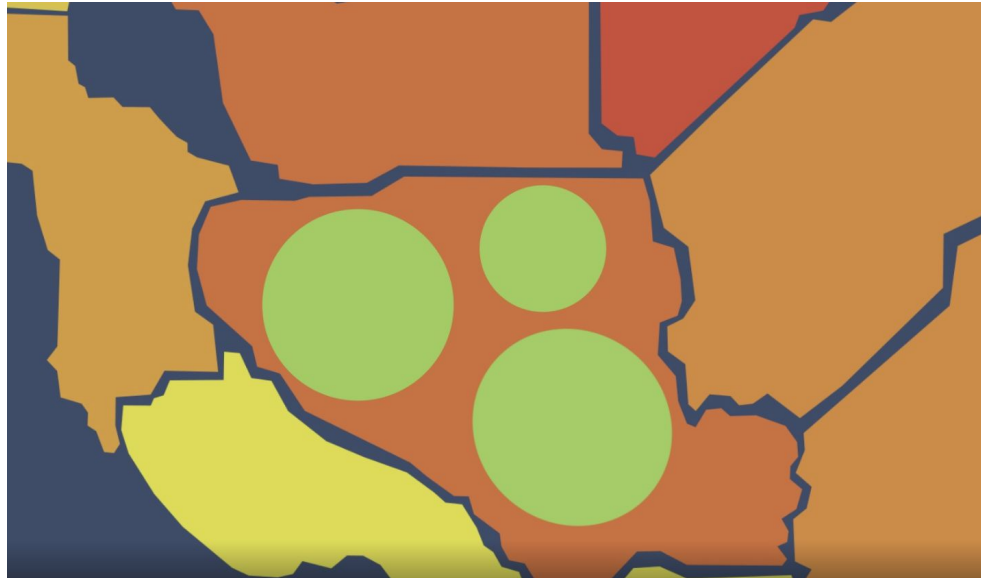
↑ Same Primers
↓ Split Purification

Fluorimetric LAMP

- qPCR Machine Readout
- 1/3 time compared to PCR
- Quantitative
- Use for immediate confirmation



High Level Pooling of Negatives

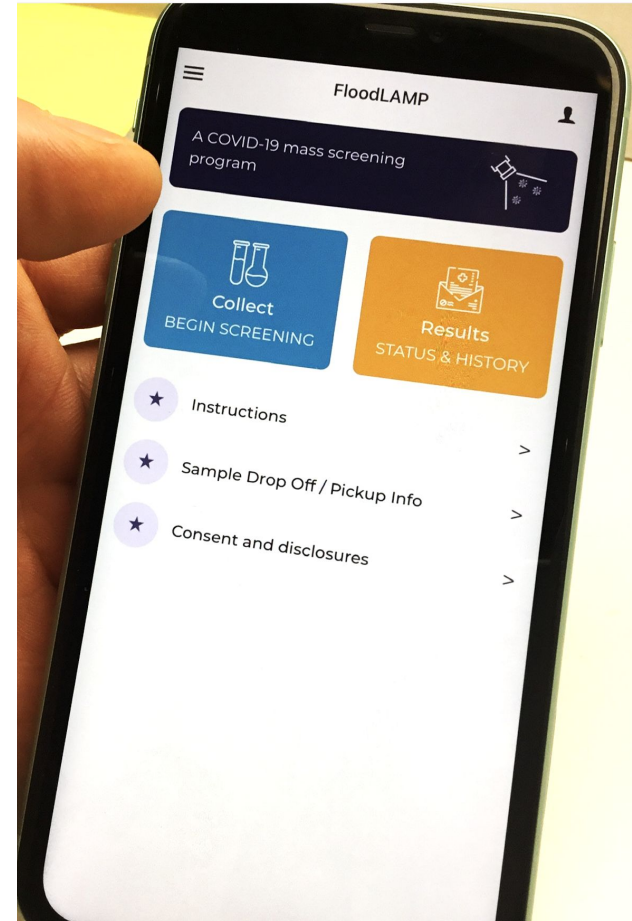
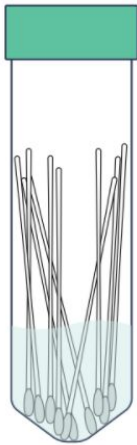


Get Green and Keep Green - [animation link](#)

Prevalence vs Pool Level tradeoffs do not apply to frequent screening.

Decentralized Pooled Collection

- At-home and on-site collection.
- Pooling by household or pod.
- Custom app for collecting samples and tracking results.



The Problem



Rapid antigen tests are not good enough

- poor sensitivity
- too many false positives and false negatives



PCR isn't scalable enough

- no centralized testing strategy
- only incremental capacity increases on horizon

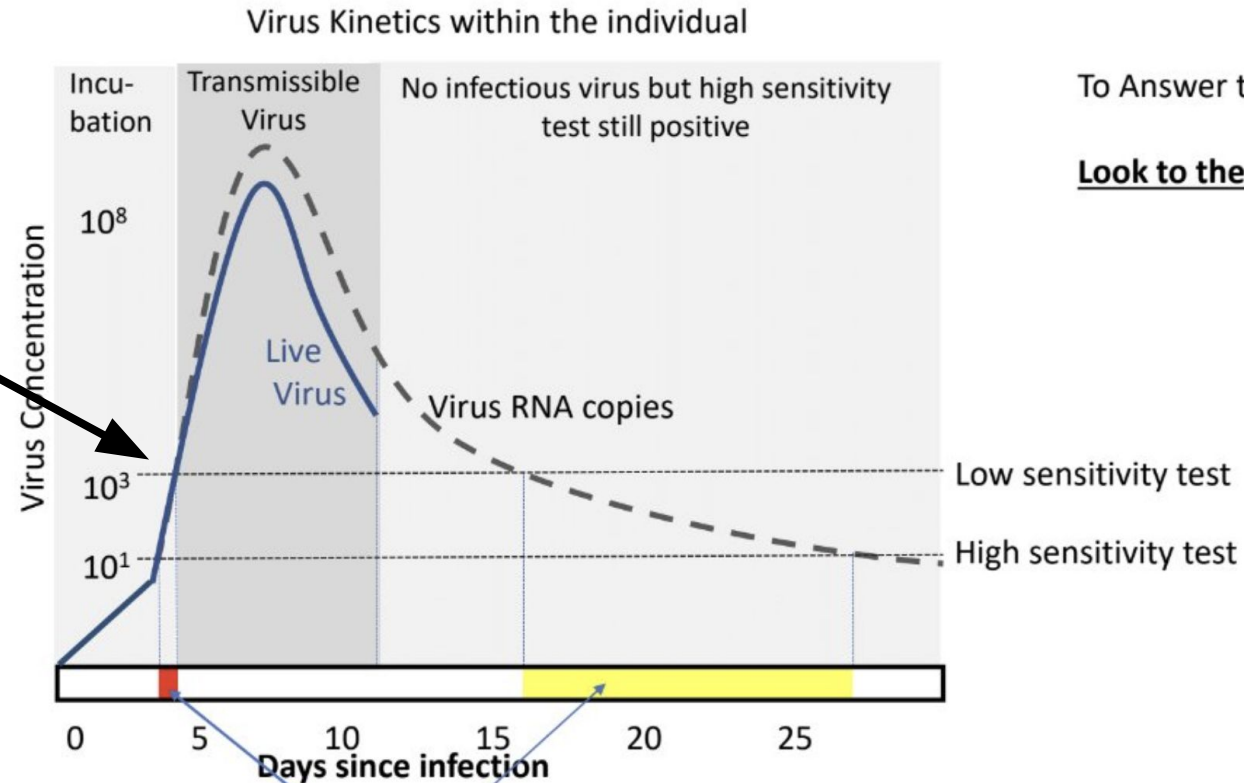
Current approaches cannot scale quickly.



Viral Load and Infectious Detection

Threshold for
"Capable to
Infect Others" ~
 $1e6/ml$

Pooling uses assay
sensitivity to scale
up coverage - and
detect unknown
infectious people
in interacting
populations



To Answer the question....

Look to the virus kinetics

Low sensitivity test *false-negative* only when taken in red/yellow times
But...Only the red area matters clinically & is a very short window of time!

Saliva Direct "most unique EUA" – FDA

- Is a protocol not a product
- Open Access - any CLIA can use
- Lots of clone assays being used

- Yale can "designate" CLIA labs

Yale SCHOOL OF MEDICINE

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SalivaDirect: What You Need to Know About the New COVID-19 Test

September 03, 2020 by Michael Greenwood

COVIDTracker

About our Test Development

SalivaDirect™ has moved!

For information regarding SalivaDirect™, including lab designation and up-to-date validation results, please visit our new page at <https://publichealth.yale.edu/salivadirect/>.

COVIDTrackerCT will remain home to our surveillance, open-source protocols, and research (including test development).

Saliva

Combine & mix

Heat inactivate

Dualplex RT-qPCR

Created with BioRender.com

FloodLAMP.bio

CC BY SA

contact@floodlamp.bio

SalivaDirect: Lab authorization request form

SalivaDirect received Emergency Use Authorization (EUA) from the Food and Drug Administration (FDA) on August 15th, 2020.

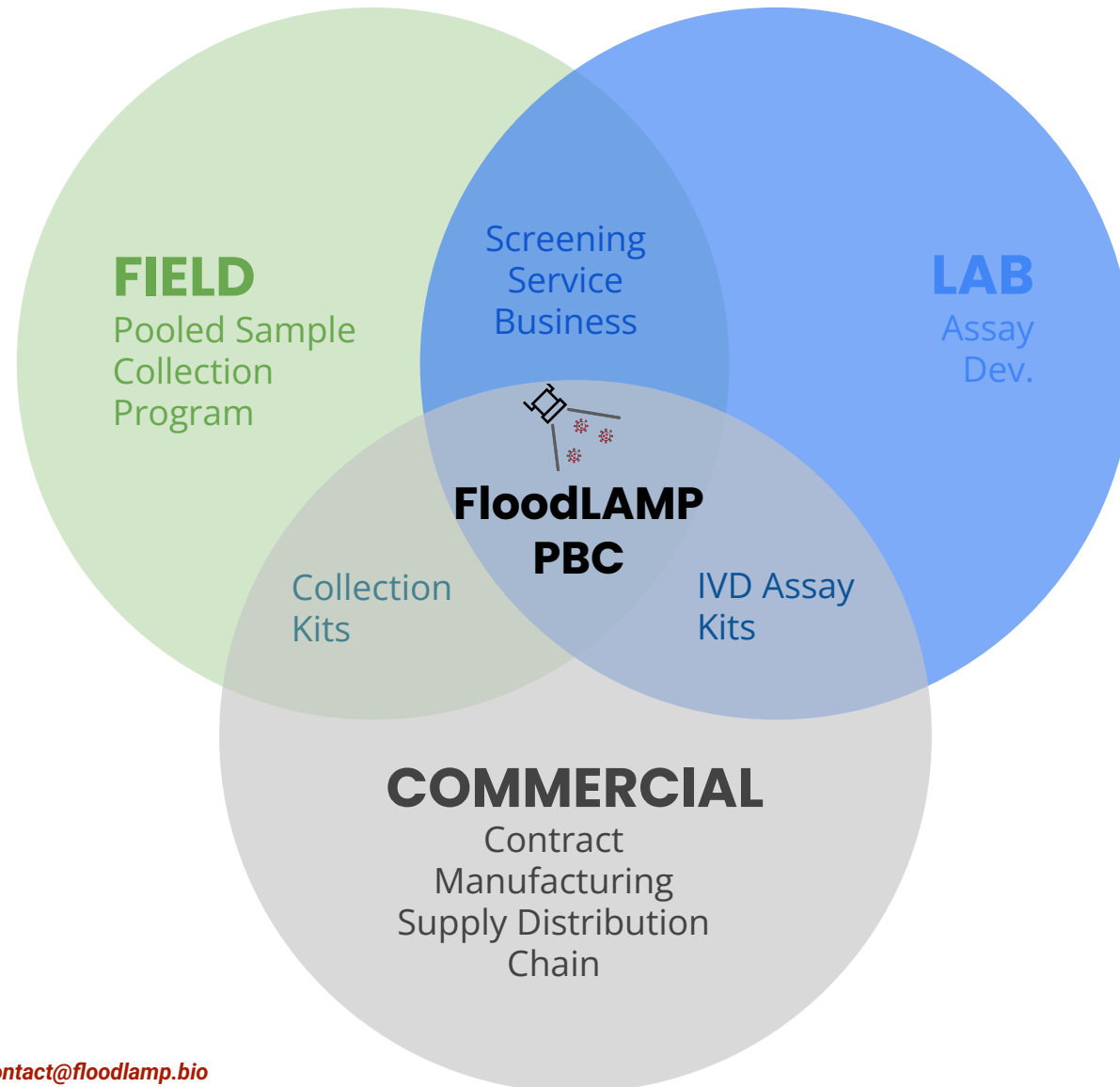
The SalivaDirect FDA EUA is for a Laboratory Developed Test, but we have the right to designate other labs its use.

Only high complexity CLIA-certified labs in the United States can become authorized to run SalivaDirect. Please fill out this form if you would like to receive more information on how to become a designated lab.

Do you have access to at least 5x SARS-CoV-2 positive and 5x negative saliva samples for CLIA verification? *

Would you be open to collaboration on bridging studies for your automation systems?

FloodLAMP PBC at Intersection



Roadmap to Ending Crisis through Testing

Expand Saliva Direct

- Create efficient designation valid. process
- Establish network of labs
- Pool to level of 5
- Has immediate impact
- Build sample collection org.

Validate Pooled Colorim LAMP

- Open assay IVD EUA
- Open pooled collection EUA
- Pooling to baseline 8-10
- Pool pools to 20-100 w/ LoD target

Massive Rollout

- Build commercial ops
- Bring in NGO's to scale
- Government Funding (DOD)



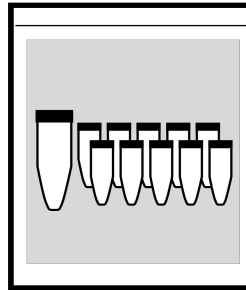
Sample Collection Workflow

1 REGISTER



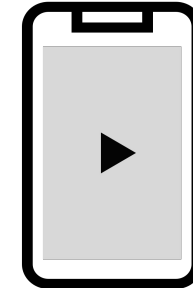
- Visit app and register
- Profile information added
- Your organization (if applicable)

2 GET



- Receive kit
- Review instructions

3 LEARN



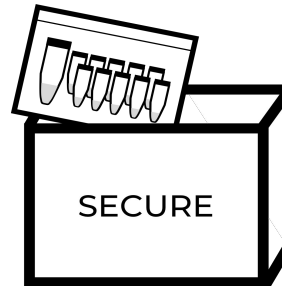
- Watch videos on collection process
- Sponsor guide and preparation

4 COLLECT



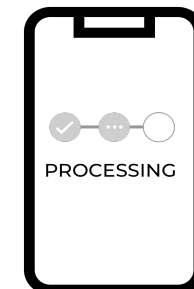
- Add collection participants in app
- Collect individual samples + record
- pool samples + record

5 DROP OFF



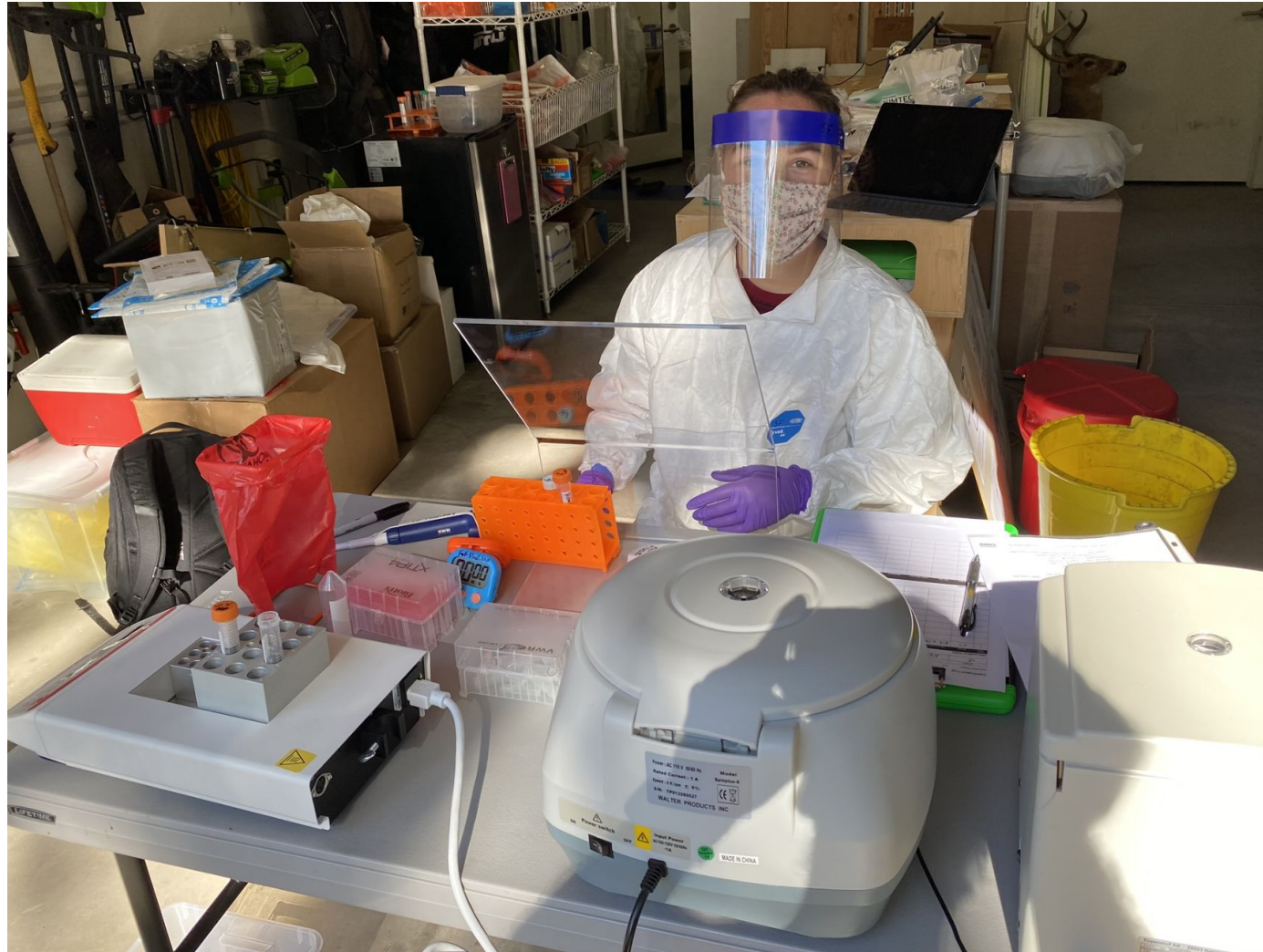
- Drop samples in secure drop box and record in app OR
- Drop with staff member who records in system

6 VIEW RESULTS



- See state of your sample (in transit, processing, result, etc)
- See updates for next collection
- Sharing options for adoption spread

Inactivation Station



Problems with Antigen Tests



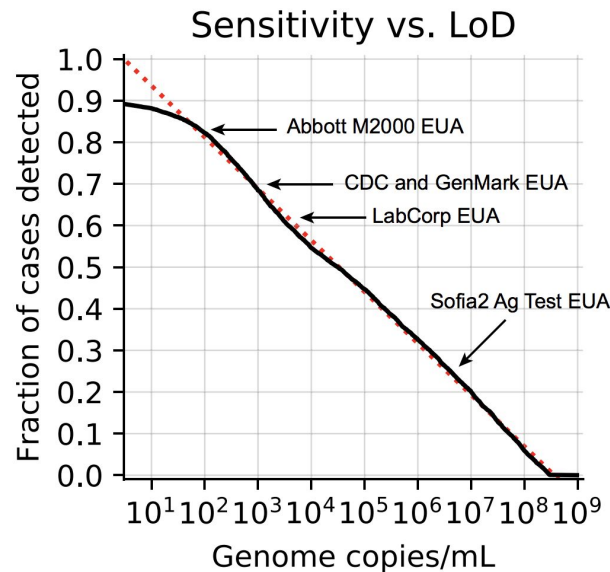
Los Angeles Times

CALIFORNIA

Coronavirus outbreaks ravage nursing homes again, even after Newsom safety measures

rapid antigen tests the federal government has sent to nursing homes recently have been “absolutely useless,” she said. They produce too many false results.

"An antigen test would **probably be an utter disaster** used twice a week, because you'd have so much community spread you might not be able to stop it" - Prof at small college testing all 5,000 students 2x per week with PCR.



FDA Explicit Priorities

"we're very eager to authorize home diagnostic tests either over-the-counter or by prescription. I have to say though we haven't received a single EUA application for a home test, not one and we've had our template out there for a long time and we've clearly expressed flexibility so we want to reiterate. We want to see home test submissions and we're willing to be very flexible here."

"So what we have attempted to do is provide as much flexibility to add an asymptomatic claim to anybody who wants one."

- Tim Stenzel, FDA Director of In Vitro Diagnostics (9-16 Town Hall)

"Perfection is the enemy of the good"

But what about RNase contamination?
What about false positive?
What about the FDA and CLIA?

